



Investment Society
Imperial College London



Lecture 6

Fixed Income

Securities Education Certificate

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Background

Some context to introduce you to a concept

Fact

You will need to recall this information in the exam

Model Example

A fully detailed breakdown of an example (question)

Try it yourself

It's your turn to give this problem a go and test your knowledge!

Equation sheet

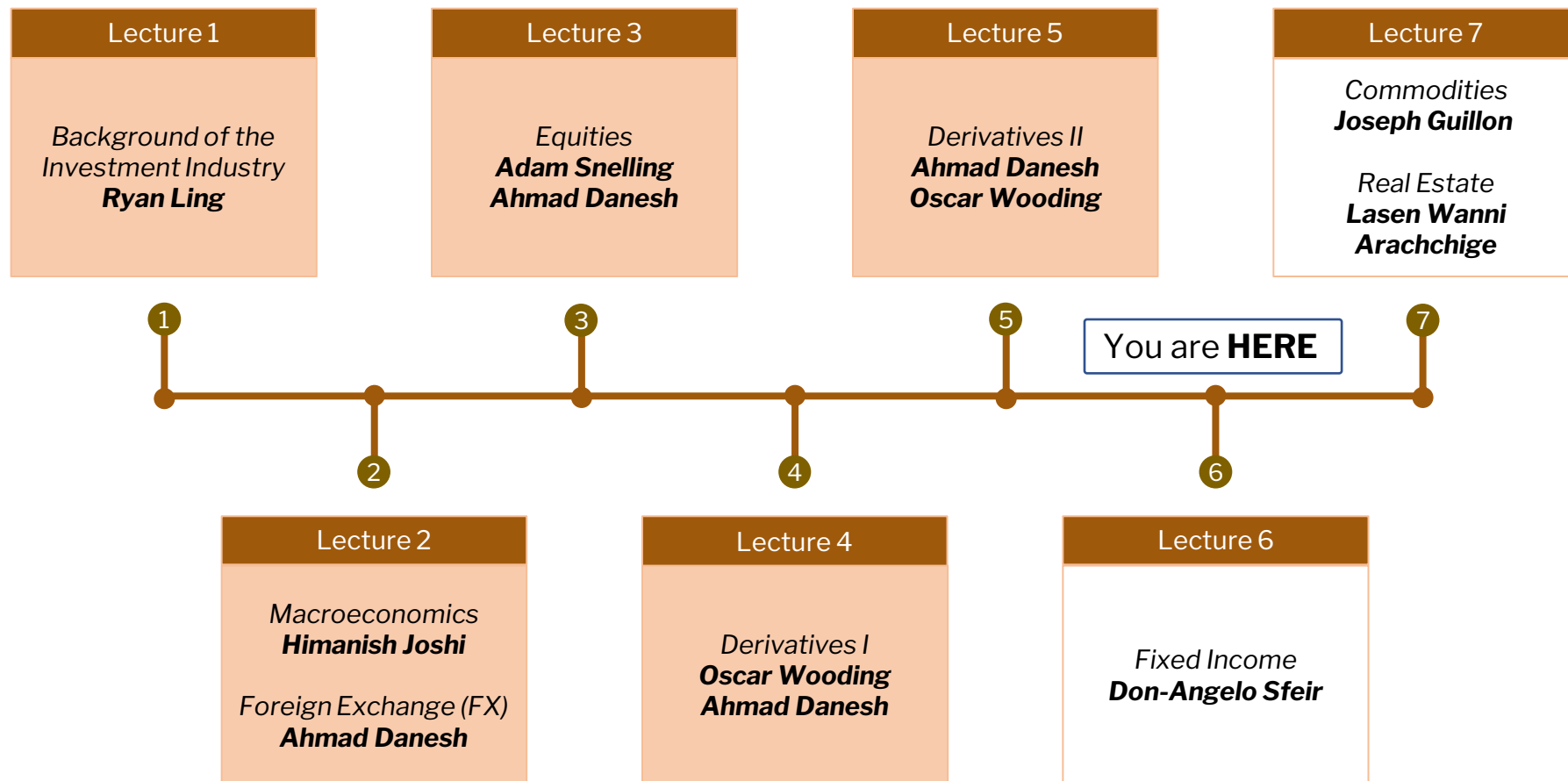
Any equations that you may need for the exam are provided

Taking it further

(non-examinable) stuff to show you where this concept can be applied

The slides are colour-coded to aid your understanding of the course and ensure that you are familiar with the demands of the content provided in each slide.

Lecture Timeline



Let's begin...



Don-Angelo Sfeir

Lecturer

Lecture 6

Fixed Income (inc. Bonds)

Recap of Bonds Primer

Is there an alternative way for a company to raise funds?

👉 There are two standard ways for a company to raise funds.

Issue **equity**

Give up ownership of future profits

Issue **debt**

Pay back original amount **WITH** interest

Is there a third way to raise funds that benefits the owners of the *company* the most?

Retain ownership of future profits

Owners set their own interest rate

Your notes

Recap of Bonds Primer

*Retaining ownership while setting your own interest rates is a **BOND**.*

👉 Bonds allow borrowers to borrow money on **THEIR** (the borrower's) terms, **NOT** the issuer's.

👉 The borrower sets the interest rate and the size of the money being loaned.

👉 The borrower does **NOT** lose any ownership (equity) in the business.

👉 An interest rate is still paid on the original money loaned.

Government Bonds

LARGE Corporation Bonds

Your notes

Recap of Bonds Primer

*Let's look at an example of a UK Government Bond (**'Gilt'**) to show how it works.*

The UK Government wants to raise £1 billion for a new project to pay back in **10 years**.

The Treasury issues bonds of **£1000** to investors → **'par' / 'face value'**

The *annualised* Interest Rate of the Bond is **3%**.

👉 Bonds are often issued during a **budget deficit** i.e. not enough money to pay for projects.

👉 Investors think the Government is a 'safe' / 'risk-free' bond issuer → Gov. Bonds = 'risk-free return'

Your notes

Recap of Bonds Primer

How do we discuss this '**gilt**' in the world of finance and investment?

The UK Government wants to raise £1 billion for a new project to pay back in **10 years**.

Bond Principal = £1000

Coupon Rate = £3%

Maturity Date = 10 years (from issue)

👉 **Bond Principal** → The amount of money a *lender* pays to the *borrower*.

👉 **Coupon Rate** → The interest paid by the *borrower* ('bond issuer') to the *lender* (or the 'bondholder')

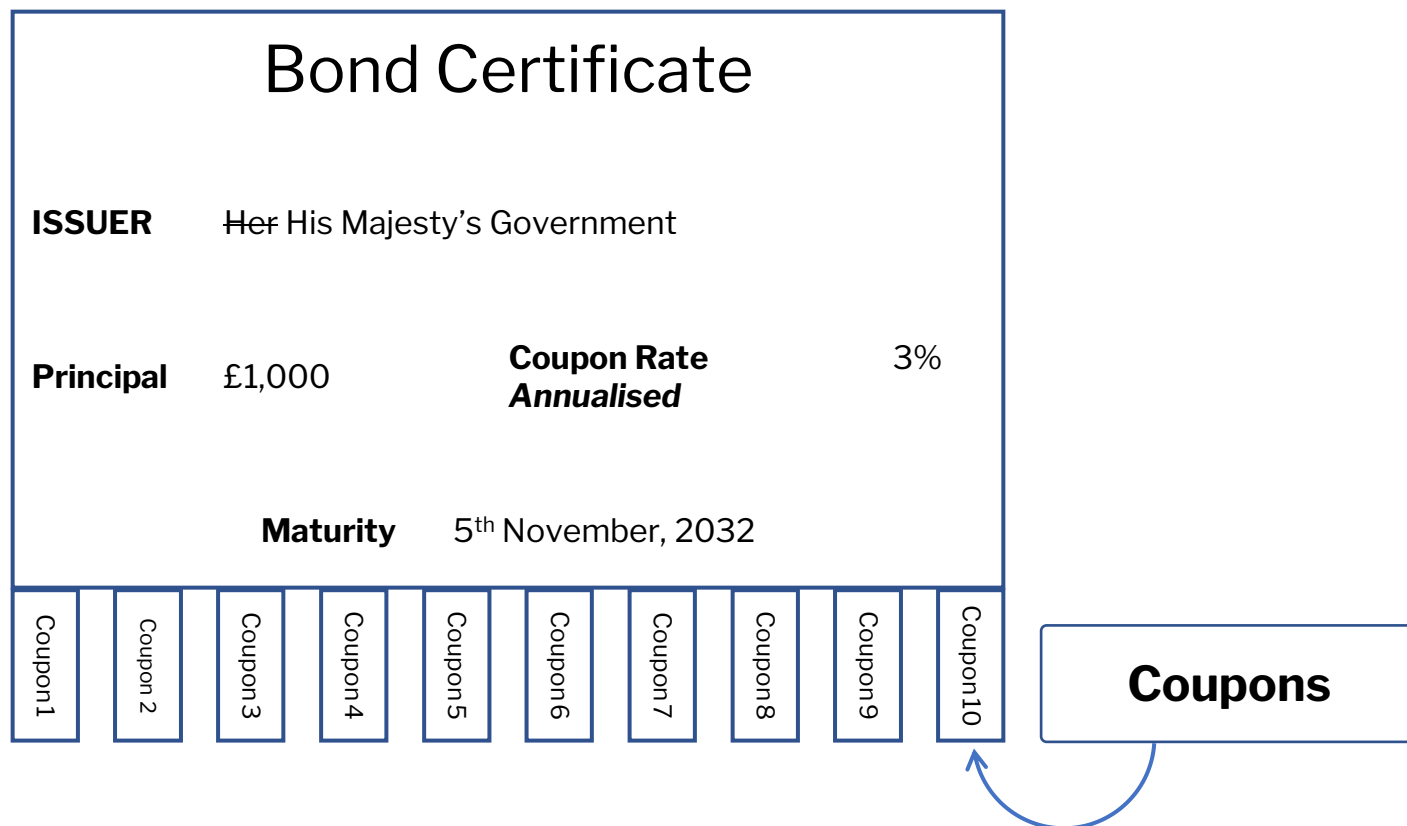
👉 **Maturity** → The date when the bond is paid off completely by the bond issuer.

Your notes

Recap of Bonds Primer

How did bonds look in the old days?

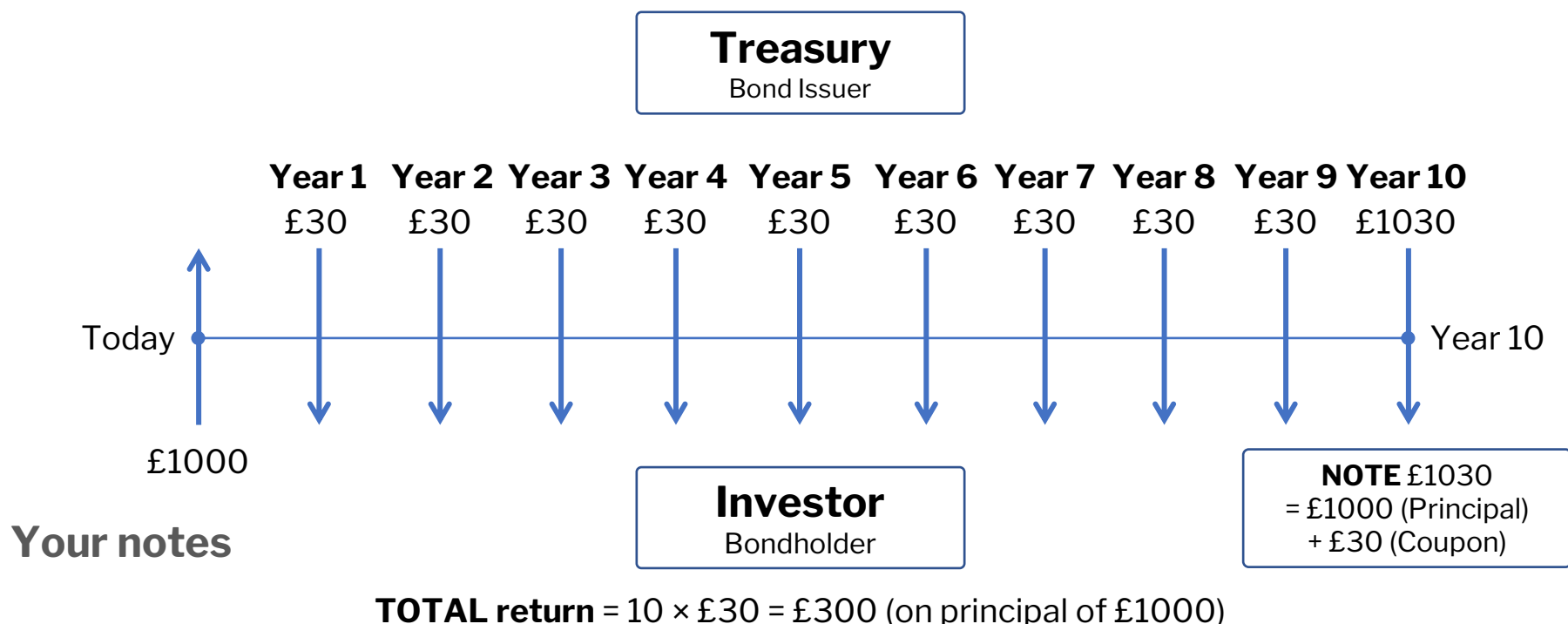
👉 Bond issuers published *certificates* for bondholders to track a bond until maturity.



Recap of Bonds Primer

You can also model bonds in terms of their **cash flow**.

👉 Bonds are just transfers of money between two parties → model the *cash flow*.



Recap of Bonds Primer

You can also calculate another property of bonds... Bond Yields!

👉 Bond **yields** show the return on a bond for an investor.

$$\text{Bond Yield (\%)} = \frac{\text{Coupon Rate}}{\text{Market Price}} \times 100$$

Why are we dividing by the *Market Price*?

👉 Bonds (c.f. stocks) can be traded on *secondary capital markets* → covers **ALL** bond types (new/old).

👉 If the price of a bond goes down, the bond **yield** goes up → inverse relationship.

Your notes

Introduction to Fixed Income

What is Fixed Income?

- ➡ An investment under which the borrower makes payments of a fixed amount on a fixed schedule.
- ➡ Fixed Income (**FI**) holds less risk than equity investments.
- ➡ FI generates a steady source of income.
- ➡ FI may also have complete downside protection.
Depending on the terms of the fixed income security.
- ➡ FI is the world's largest asset class ca. \$200T globally.
Almost 3x the size of combined global equity market valuation.

Bonds

Let's look at the most basic fixed income... a bond. Its structures and uses vary.

Sovereign Bonds

A debt security issued by a national government to raise money.

Financing
government
programs

Paying interest
on current debt

Other
government
spending needs

Lowest investment risk → 'risk-free'

Can be denominated in a foreign currency
OR the government's domestic currency.

Corporate Bonds

A debt security issued by a corporation in order to raise debt.

Regular
company
operations

Refinancing
debt

Financing M&As

Can carry more risk than government bonds.

Spectrum of bond *qualities*.

Your notes

Bond Characteristics

Bonds may be different in the fine print but they share similar characteristics.

👉 The bond's **maturity** is the date when the principal of the bond is paid back to bond holders.

👉 The company's bond obligation ends → Maturity defines the *lifetime* of the bond.

👉 Investors look at a bond's maturity to assess its investment potential.

Short Term

Medium Term

Long Term

Maturity in 1 – 3 Years

Maturity in ~ 10 Years

Maturity > 10 Years
Longer Time Periods

Your notes

Bond Characteristics

How is cash flow guaranteed if the issuer fails to pay the bond back to the holder?

Secured/Unsecured

Pledges assets to bondholders if the issuer cannot repay the obligation i.e. **collateral**

e.g. a mortgage-backed security (MBS)

Issuer's assets are used to repay debts
in cases of default

Liquidation Preference

During bankruptcy, investors are repaid in the order of the **capital stack**.

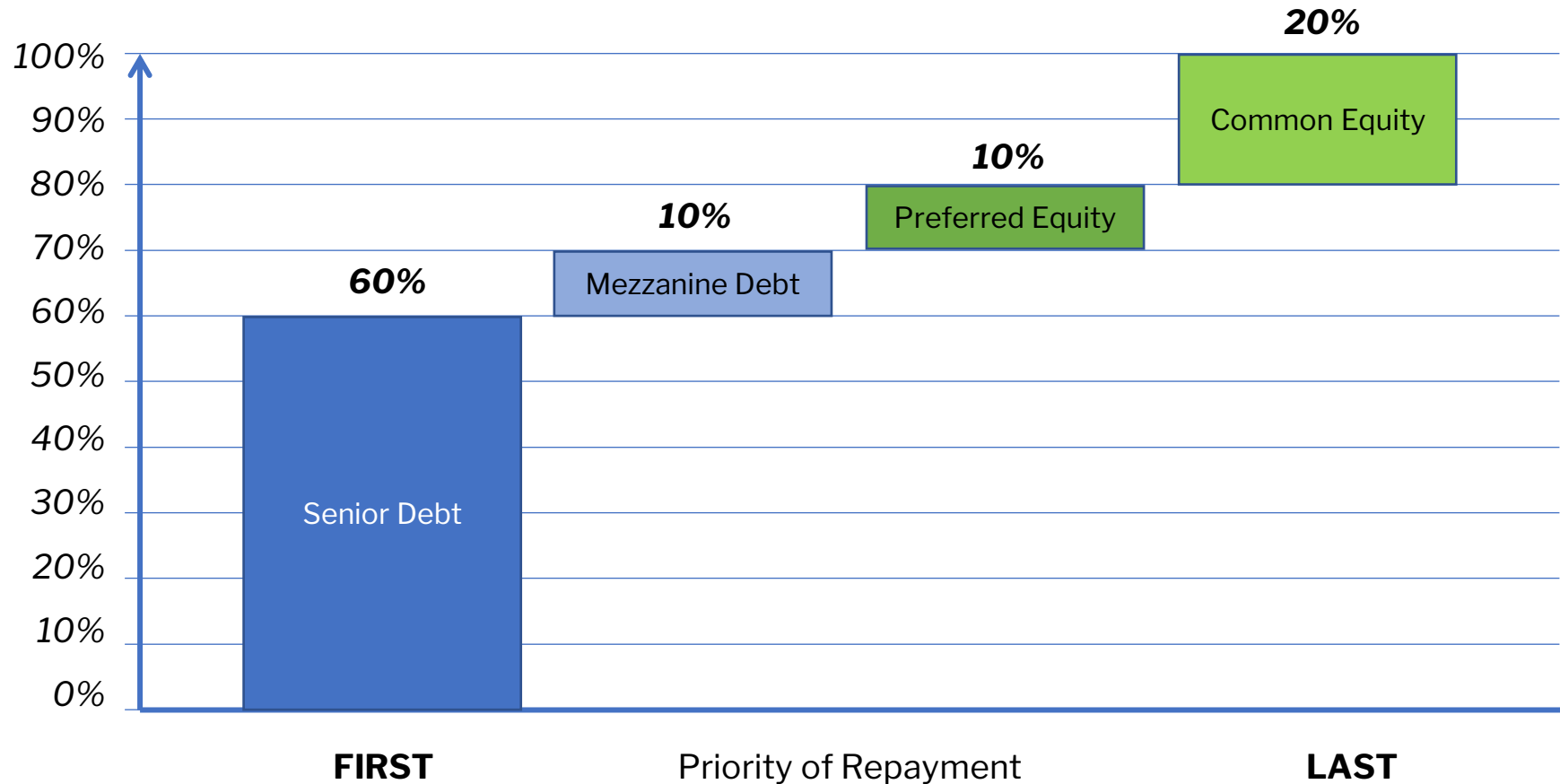
Senior Debt > Junior Debt > Equity

Bonds tend to be Senior Debt

☞ Paying off investors during bankruptcy is called **liquidation**.

Bond Characteristics

There's an order in which debts and equities are paid off during liquidation.



Coupon Rate

The coupon amount represents the interest paid to **bondholders**.

$$\text{Coupon Rate} = \frac{\text{Annual Coupon}}{\text{Par Value of Bond}}$$

☞ Coupons are (usually) paid out annually or semi-annually.

☞ Coupons qualify a bond as a Fixed Income asset.

☞ Coupons are similar to stock dividends.

☞ Coupon rate is an underlier for various yields.

Coupon Rate

Let's look at an example of how to evaluate coupon rates.

$$\text{Coupon Rate} = \frac{\text{Annual Coupon}}{\text{Par Value of Bond}}$$

Meta plans to release a corporate bond to get through the tech crash.

The proposal is a \$1500 bond that pays a yearly coupon of \$60

What is the coupon rate of the bond?

$$\frac{\$60}{\$1500} = 0.04 = 4\%$$

Bond Characteristics

Bonds can vary e.g. corporate vs government, along with callability. How?

Tax status

Corporate bonds are usually *taxable*.
Government bonds are usually ***tax-exempt***.

Tax-exempt bonds allow for untaxable income.

Tax-exempt bonds normally have a lower coupon rate than equivalent, taxable bonds.

You calculate the tax-equivalent yield to compare the return of tax-free bonds with taxable bonds.

Callability

Call provisions allow bonds to be paid off at earlier dates.

(Usually) at a slight premium to the principle.

Bonds with a call option
tend to have higher coupon rates.

A company can *call* its bonds if they can borrow at lower interest rates.

Your notes

Risks of Bonds

Bonds are relatively safe investments, but they do come with certain risks.

Prepayment

Companies want to pay off bonds early if interest rates fall.

Investors are left to reinvest funds in a lower interest rate environment.

Interest Rate

If interest rates rise, investors will be stuck with bonds yielding below market rates.

The greater the time to maturity, the greater the interest rates risk.

Credit/Default

A firm can default on its debt → the issuer no longer pays off the bond.

A firm has greater operating income and cash flow compared to its debt. This is called **safety**.

Bond Ratings and Agencies

Bonds are normally rated so investors to judge their worthiness for a strategy.

Agencies

They assign credit *ratings* to bonds.
See right for examples.

Investment grade bonds are graded Aaa to Baa.

Junk bonds are graded Ba to below.

Junk bonds may offer higher coupon rates.

Firms may not have their bonds rated.
The investor judges their bond quality.

Here are the 'Big 3' ratings agencies...

S&P Global
Ratings

MOODY'S

FitchRatings

'Big 3' account for ~95% of credit ratings.

Bond Ratings and Agencies

Let's look at these ratings in more detail...

Grade	Description	Category
Aaa	Prime	Investment-grade Unlikely to default Tend to remain stable investments
Aa1	High grade	
Aa2		
Aa3		
A1		
A2	Upper medium grade	
A3		
Baa1		
Baa2	Lower medium grade	
Baa3		
Ba1		
Ba2	Non-investment grade speculative	
Ba3		
B1		
B2	Highly speculative	
B3		
Caa1		
Caa2	Substantial risks	
Caa3	Extremely speculative	
Ca	Default imminent with little prospect for recovery	
C		In default

Bond Yields

Bond yields are measures of return → **Yield to maturity** is one example.

$$YTM = \sqrt[n]{\frac{\text{Par Value}}{\text{Purchase Price}}} - 1$$

👉 Yield to maturity (YTM) measures the return on a bond.

👉 YTM is valid only if two conditions are **BOTH** met:

The bond is held to maturity

All coupons are reinvested into the bond

👉 YTM is the most widely used yield measurement.

Your notes

Yield to Maturity

Let's look at an example of how to use YTM.

$$YTM = \sqrt[n]{\frac{\text{Par Value}}{\text{Purchase Price}}} - 1$$

An investor wants to know the yield to maturity of five-year US treasury bonds

10 bonds were bought on issuance at \$100 per bond

Every reinvestment of the annual coupon of \$10 was at a discounted value of \$95.

What is the YTM of this investment?

Your notes

Yield to Maturity

An investor wants to know the yield to maturity of five-year US treasury bonds.

If 10 bonds were bought on issuance at \$100 per bond, and every reinvestment of the annual coupon of \$10 was at a discounted value of \$95, **what is the YTM of this investment?**

$$YTM = \sqrt[n]{\frac{\text{Par Value}}{\text{Purchase Price}}} - 1$$

$$YTM = \sqrt[5]{\frac{1000}{1000}} - 1 + \sqrt[4]{\frac{100}{95}} - 1 + \sqrt[3]{\frac{100}{95}} - 1 + \sqrt[2]{\frac{100}{95}} - 1 + \frac{100}{95} - 1$$

$$YTM = 0 + 0.0129 + 0.0172 + 0.0260 + 0.0526$$

$$YTM = 0.1087 = 10.87\%$$

Your notes

Yield to Maturity

Let's try another example of how to use YTM.

$$YTM = \sqrt[n]{\frac{\text{Par Value}}{\text{Purchase Price}}} - 1$$

An investor wants to know the yield to maturity of three-year US treasury bonds

10 bonds were bought on issuance at \$150 per bond

Every reinvestment of the annual coupon of \$15 was at a discounted value of \$125.

What is the YTM of this investment?

Your notes

Yield to Maturity

An investor wants to know the yield to maturity of three-year US treasury bonds.

If 10 bonds were bought on issuance at \$150 per bond, and every reinvestment of the annual coupon of \$15 was at a discounted value of \$125, **what is the YTM of this investment?**

$$YTM = \sqrt[n]{\frac{\text{Par Value}}{\text{Purchase Price}}} - 1$$

$$YTM = \sqrt[3]{\frac{1500}{1500}} - 1 + \sqrt[2]{\frac{150}{125}} - 1 + \frac{150}{125} - 1$$

$$YTM = 0 + 0.0954 + 0.02$$

$$YTM = 0.2954 = 29.54\%$$

Your notes

Bond Yields

There are two types of yields that you might see. **What are their differences?**

Current Yield

$$\text{Current Yield} = \frac{\text{annual coupon income}}{\text{Current Price}}$$

Compare the bond's interest income to the dividend income provided by a stock.

It ignores possible capital gains or losses associated with the principle.

Current yield is most useful for investors concerned with current income only.

Nominal Yield

$$\text{Nominal Yield} = \frac{\text{annual coupon income}}{\text{principal}}$$

Simply the percentage of interest to be paid on the bond periodically i.e. the coupon rate

N.B. it does not estimate return accurately unless the current bond price is the same as its principal value.

Nominal yield is used only for calculating other measures of return.

Your notes

Bond Yields

Other yield parameters are useful too! So, let's check two of them out.

Yield to Call, **YTC**

$$\text{Bond Price}_{PV} = C \times \frac{1 - \frac{1}{(1+r)^n}}{r} + \frac{P}{(1+r)^n}$$

C is the Coupon Rate, **n** is the time periods until call, **r** is the Yield to Call (YTC) & **P** is the bond Principal.

Investors receive a slightly higher yield if the called bonds are paid off at a premium.

An investor calculate the YTC to see whether the prepayment risk is worthwhile.

(Annualised) Realised Yield, **ARY**

$$ARY = \frac{C + \frac{P_F - P_0}{n}}{\frac{P_F - P_0}{2}}$$

P_F is the final bond price, **P₀** is the initial bond price (at purchase) & **n** is the years that the bond is held until sale.

Realised yields should be calculated only if the bond is held for a certain period.

Future bond price must be estimated for the calculation.
→ **ARY** is an *estimation* of return.

These formulae may be required in the exam – they will be given to you.

An intuitive mathematical derivation of the YTC and ARY is given in the notes.

Yield Curves

*You can plot bond yields for the same bond with varying lifetimes → **yield curves**.*

👉 Bonds with longer durations until maturity offer greater bond yields.

👉 The bondholder is compensated for exposure to more risk over a longer time period.

👉 Yields of the same bond *type* but with different lifetimes can be plotted.

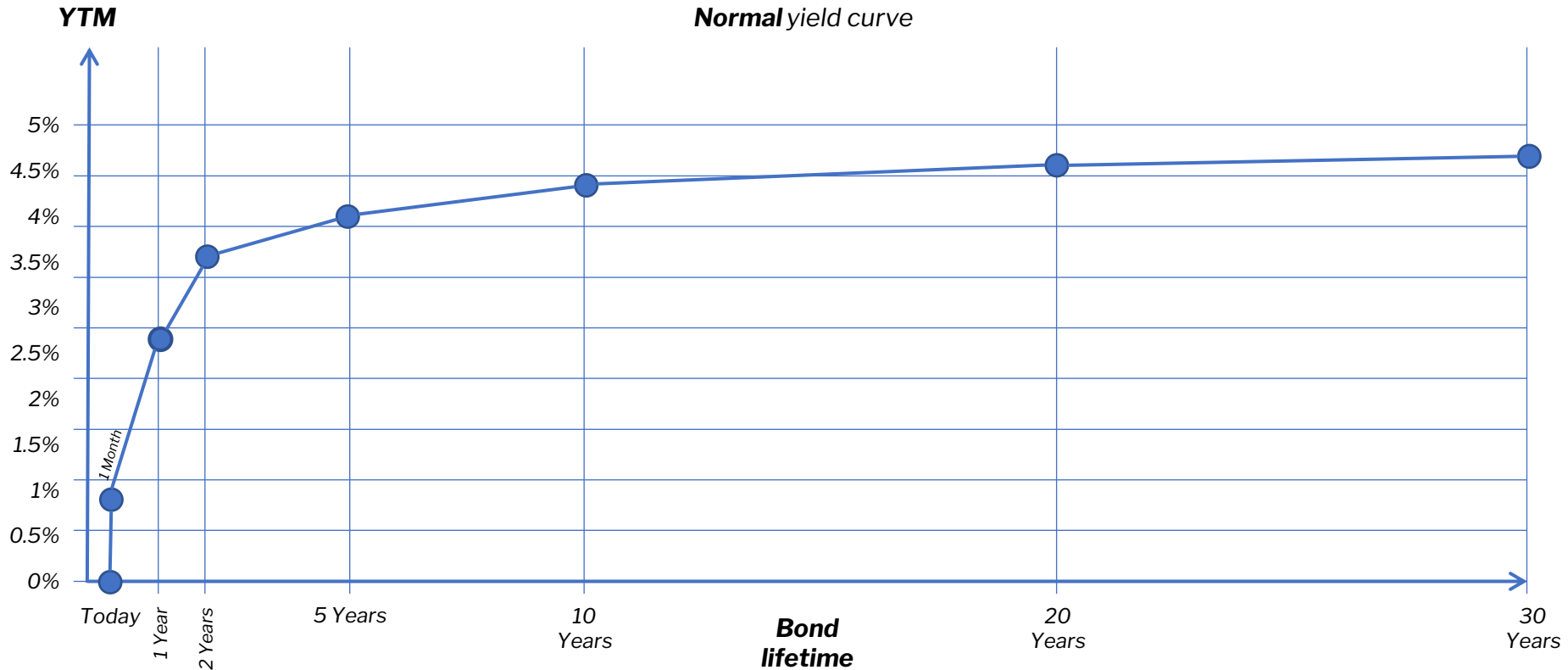
👉 The resulting graph is known as a **yield curve**.

Yield curves are **HIGHLY** important in Fixed Income!

Your notes

Yield Curves – UK Gilt

What about the most risk-free bond you can find... A **normal** (treasury) yield curve.

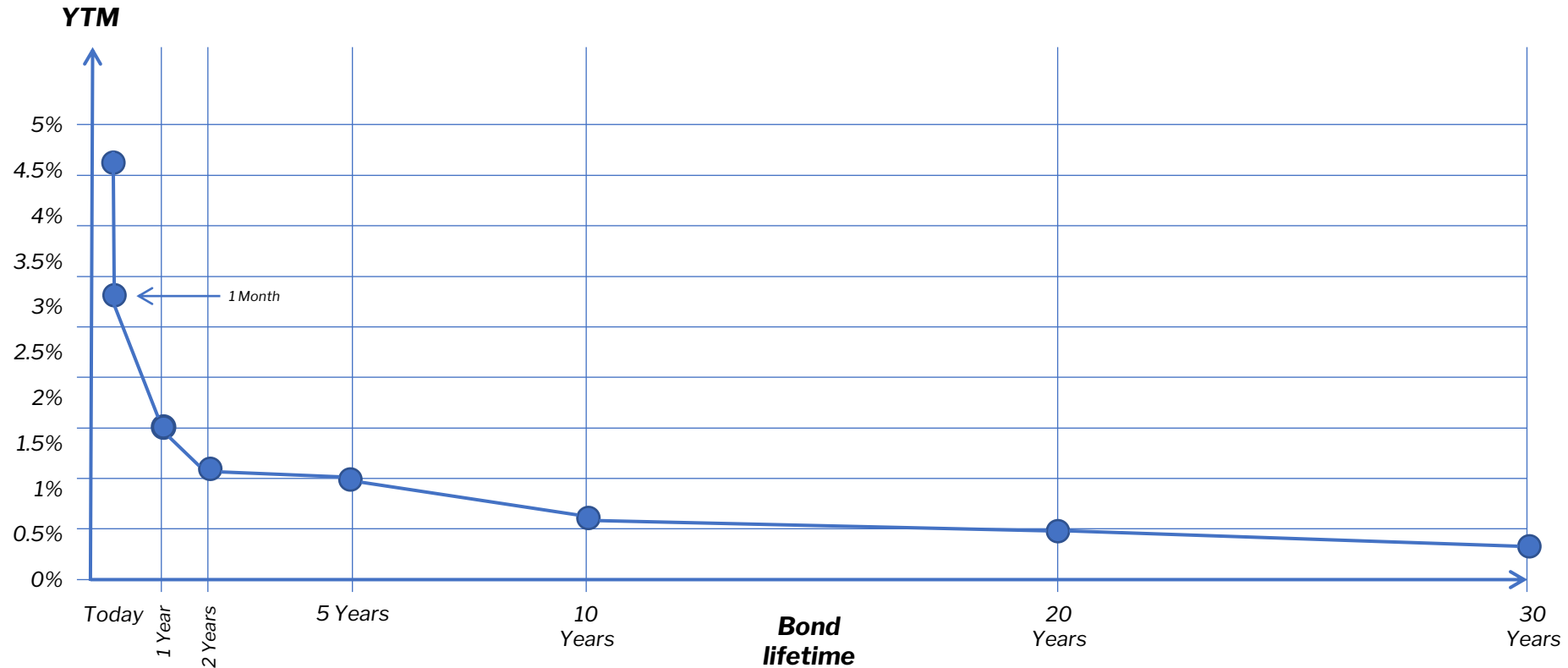


Longer bond lifetimes offer greater YTM.
This yield curve is a **normal** yield curve.

N.B. Lifetime from bond *issue*
Data only illustrative.

Yield Curves – UK Gilt

Sometimes, yield curves can slope downwards instead... → **inverted** yield curves!



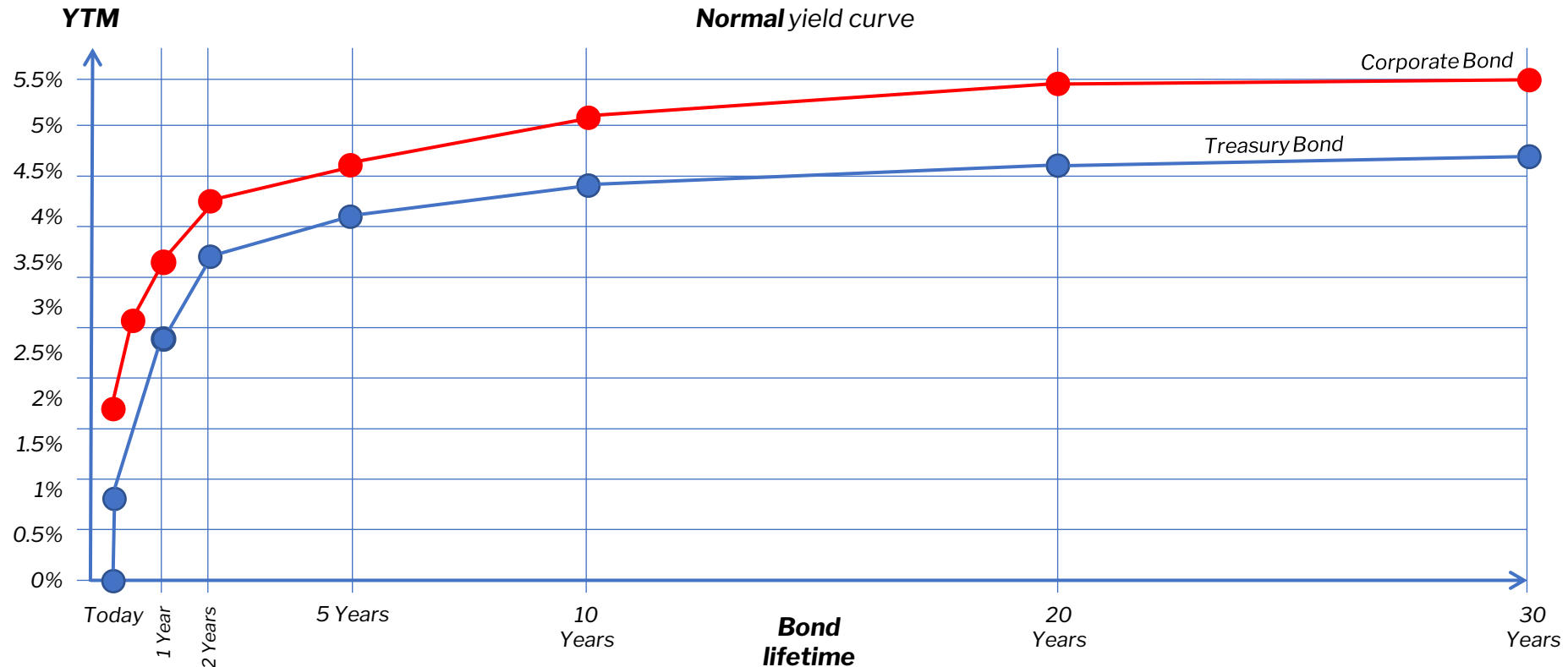
Inverted Yield curves indicate a recession.

A government bond is riskier to invest in **TODAY** than in 30 years time.

N.B. Lifetime from bond issue
Data only illustrative.

Yield Curves – Corporate Bonds

How do corporate bonds compare to treasury bonds e.g. US T-bills or UK Gilts?



👉 Corporate bonds *tend to* offer higher YTM than Treasury bonds.
You see corporate yield curves above treasury yield curves.

Yield Curves – Comparing different bonds

How do we quantify the **difference** between two types of bonds?

👉 You can use **bond spreads**.

👉 Bond spreads tell you various differences between a corporate bond and a risk-free Treasury bond.

👉 You are expected to know 2 types of spreads in the SEC...

G-spread

Z-Spread

👉 **N.B.** the treasury yield curve is called the *benchmark yield curve*.
Because treasury bonds have the lowest YTM of any bond.

*I-Spreads and OAS also exist but are beyond the scope of the course. **Please enquire if you are interested.***

Your notes

G-Spreads

So, what are G-Spreads?

☞ G-Spreads quantify the YTM difference between a corporate bond and a treasury bond.

☞ **IF** the time to maturities don't match, interpolate the treasury bond yields.

G-spread stands for **Government** Spread!

☞ G-spreads represent the *default risk* of investing into the corporate bond.

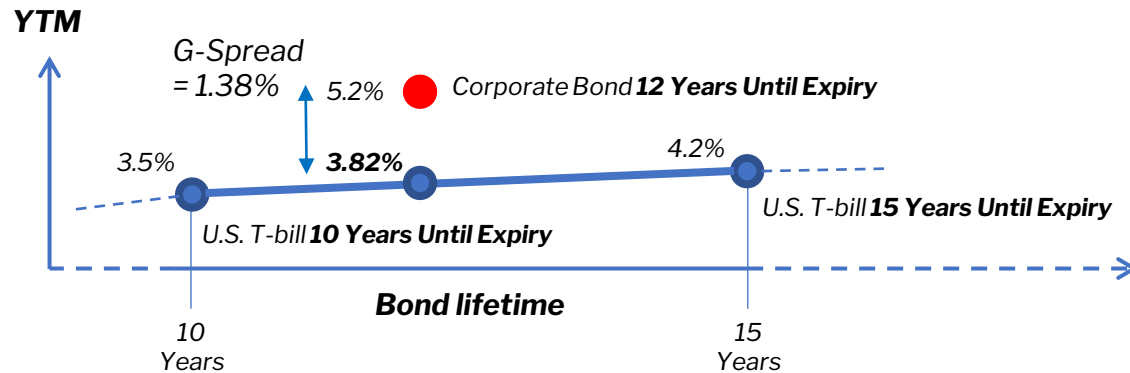
G-Spread = $Y_c - Y_g$ where Y_c is the non-treasury bond Yield and Y_g is the treasury bond Yield.

Your notes

This discussion is useless without an example. So, let's look at one!

G-Spreads

Let's consider a corporate bond that expires at a different time to the T-bill.



Interpolate the U.S. T-bill for its YTM at 12 years **to match the corporate bond...**

$$12Y YTM_{T-Bill} = 3.5 + \frac{2}{5} \times 0.7 = 3.78\%$$

$$\text{G-Spread} = 5.2 - 3.78 = 1.42\%$$

Z-Spreads

Z-spreads are a much wilder genus of bond spreads.

👉 **Z-Spreads** help discount the cash flows of a bond to the market price.

👉 Academically, it is defined as *‘the spread that must be added to each spot interest rate to cause the present value of the bond cash flows to equal the bond’s price.’*

The Z- stands for Zero-Volatility.

👉 It requires a zero coupon treasury bond i.e. a bond with no coupons!

👉 The spread is solved iteratively i.e. it cannot be solved algebraically (or ‘analytically’)

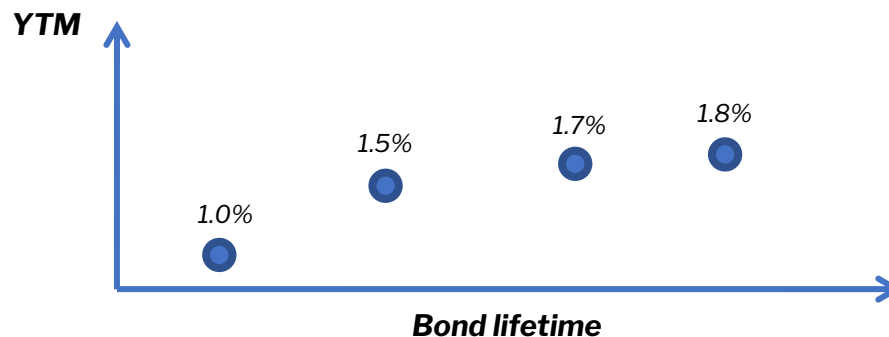
Again, this discussion is useless without an example. So, let’s look at one!

Your notes

Z-Spreads

Let's consider a corporate bond that expires at a different time to the T-bill.

👉 **YTM** curves for zero-coupon bonds are known as *spot curves*.



You can use spot rates as discount factors.

👉 These treasury rates will price the PV of a corporate bond **HIGHER** than the market price.

Consider a 4Y 10% £100 corporate bond that's trading at £127. Calculating the present value...

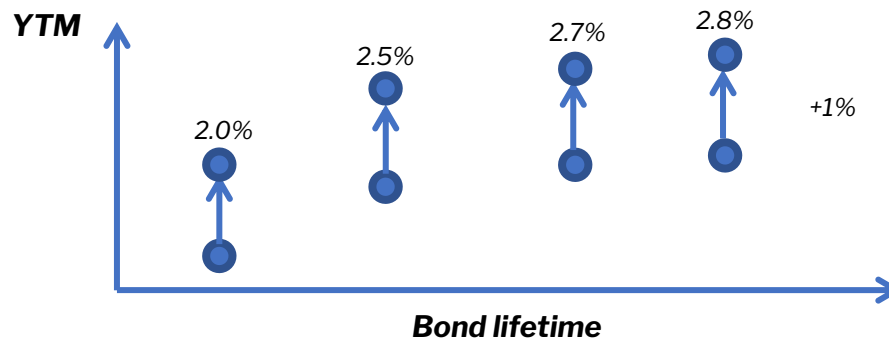
$$\text{PV Price of a 4Y 10\% 100 annual corporate bond} = \frac{10}{1+1\%} + \frac{10}{(1+1.5\%)^2} + \frac{10}{(1+1.7\%)^3} + \frac{110}{(1+1.8\%)^4} = \text{£131.53}$$

Discounting with the spot rates on the spot curve is too low!

Z-Spreads

If we add a constant to the spot rates, we could find the right discount factor.

👉 **YTM** curves for zero-coupon bonds are known as *spot curves*.



We've added 1%
to the spot curves.

This 1% is the **Z-Spread**

👉 The addition of a Z-spread will match the *future* cash flows to the *current* bond price.

$$\text{PV Price of a 4Y 10\% 100 annual corporate bond} = \frac{10}{1+2\%} + \frac{10}{(1+2.5\%)^2} + \frac{10}{(1+2.7\%)^3} + \frac{110}{(1+2.8\%)^4} = \text{£127.05}$$

The new discount factors give a bond price much closer to the market price of £127!

Z-Spreads

How does this make sense mathematically?

👉 Originally, we were calculating the present value of the bond using spot rates:

$$\text{PV Price of a 4Y 10\% 100 annual corporate bond} = \frac{10}{1+r} + \frac{10}{(1+r)^2} + \frac{10}{(1+r)^3} + \frac{110}{(1+r)^4}$$

👉 By increasing the total discount factor, the present value matched the current market value.

$$\text{PV Price of a 4Y 10\% 100 annual corporate bond w/ Z-spread} = \frac{10}{1+r+Z} + \frac{10}{(1+r+Z)^2} + \frac{10}{(1+r+Z)^3} + \frac{110}{(1+r+Z)^4}$$

👉 This constant factor is the **Z-spread**.

We solve for the Z-spread *iteratively* i.e. numerically.

👉 The Z-spread causes the spot rate (curve) to be shifted parallel up the yield curve.

Why Bonds over Equities

Generally, investing in *debt* is relatively safer than *equity* but returns do differ.

Risk

Bonds from developed governments (Treasury bonds) are considered 'risk-free'.

In 2020, a 30-year US treasury bond yielded an interest rate of about 1.7%

The current yield of the same 30 year US treasury bond is about 3.79%

More predictable returns

Bonds outperform stocks at certain times in the economic cycle.

In certain situations, people may need security and predictability.

Better than the Bank

The interest rates on bonds are greater than the deposit rates paid by banks.

Bonds will give you a relatively better return without posing too much risk.

Bonds *generally* are the best investment tools for **capital preservation**.

Interest and Bonds

The current state of the world has a large effect on bonds.

To control inflation,
Governments raise
interest rates.

Bond prices are
inversely proportional
to interest rates

Rising interest rates
raises the cost of
capital.

*2022 has seen unprecedented
soars in inflation across the
globe.*

*We are experiencing unusually
high yields in the bond market.*

Other investment tools
become less attractive due to
rising costs and higher risks

👉 Occasionally, yields can *spike* to very high levels (due to the underlying bond price crashing).

Most recently, the UK's 10-year Gilts saw yields spike tremendously.